

Name: S. Nikhileswar Reddy

Reg. No: 192525096

Course Code: CSA1613

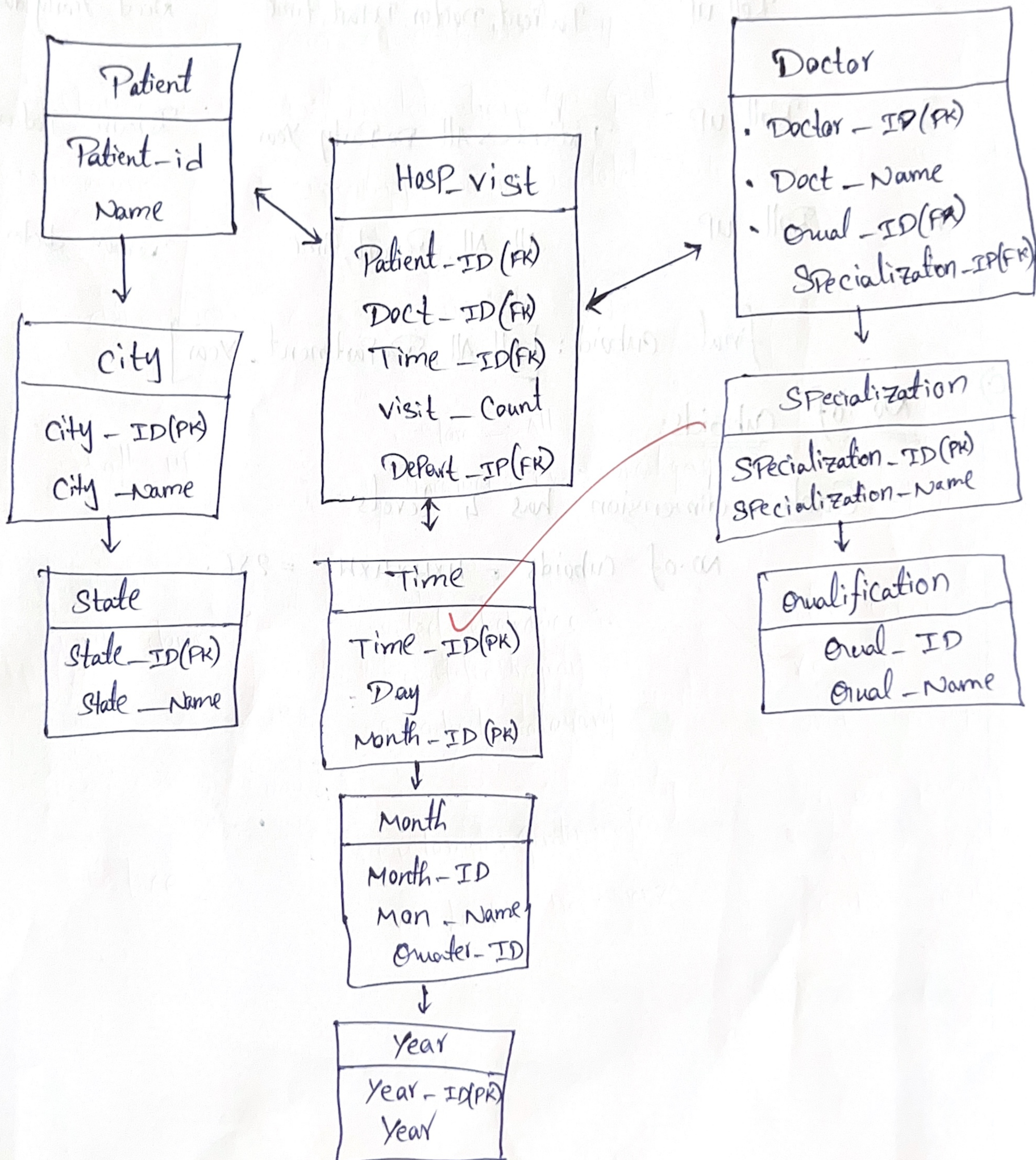
Course Name: Data warehousing
and Data Mining.

Assessment Tool-1

Co-1

1) City Hospital Data warehouse:

a) Snowflake Schema:



b) OLAP Operations:

Step	Operation	Resulting cuboid	Explanation
1	Start	Patient, Doctor, Department	Base cuboid
2	Roll up	Patient, Doctor, Depart, time	Need yearly data
3	Roll up	Patient $\rightarrow$ All Department Year	Remove Patient
4	Roll up	All, All, Depart, time	Remove doctor

final cuboid: [All, All, Department, Year].

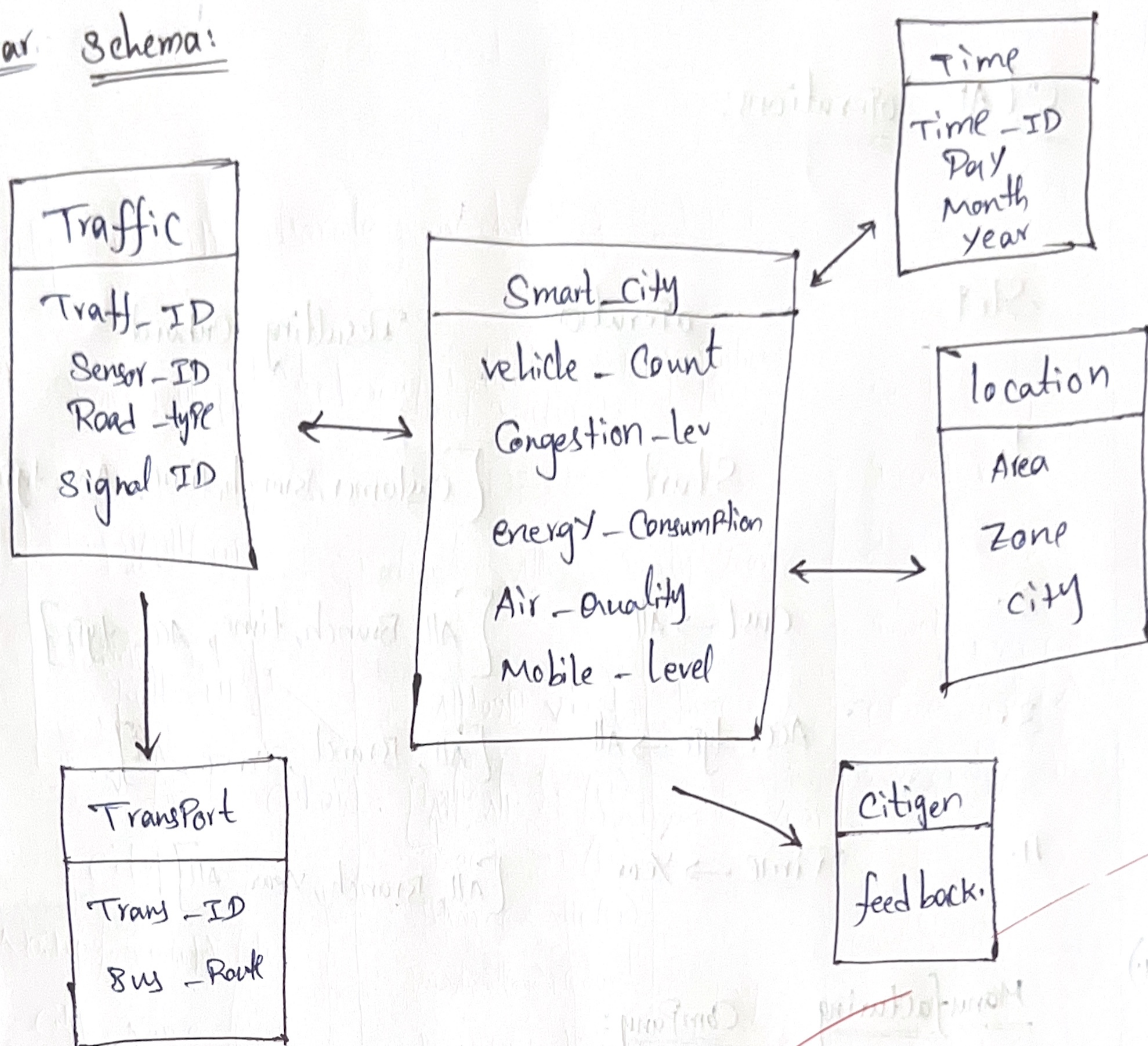
c) No. of Cuboids:

Each dimension has 4 levels

$$\text{No. of Cuboids} = 4 \times 4 \times 4 \times 4 = 256.$$

Smart city data warehouse:

Star Schema:



Justification:

- * Traffic Dimension: helps to identify Congestion Patterns
- * Location Dimension: Compress data across different Zones.
- * Time Dimension: enables analysis of daily, monthly, Yearly Trends.

3.)

Banking data warehouse:

OLAP operations:

Step	Operation	Resulting Cuboid
1.	Start	[Customer, Branch, time, Acc-type]
2.	Cust $\rightarrow$ All	[All, Branch, time, Acc-type]
3.	Acc-type $\rightarrow$ All	[All, Branch, time, All]
4.	Time $\rightarrow$ Year	[All, Branch, year, All]

4.)

Manufacturing

Company:

- * Product : Category, type, Batch
- * factory : Location, Depart, line number
- * Time : Year, Quarter, month, week
- * Supplier : Region, Rating, Contract type.

Retail chain:

Dimensions:

- * Product : Category, Sub Category, SKU
- * Store : Region, City, Store
- * Time : Year, Quarter, Month, Day
- * Customer : Membership, Tier, Age, Gender
- * Measure : Inventory Turnover, total States Revenue, units sold.

Operation

How to Perform

Result

Roll up

- Store → All
- Product : SFR → subcategory

⇒ Inventory turn
⇒ over by product Category.

Drill-down

- select electronics in Scientific region
- Product : category

• Sales Revenue
- %e Sub-category and SKU region.

Slice

Customer : Gender, male
Membership tier = Bronze

• States revenue for
BRONZE male
Customer across all stores.